



Data Analytics in Agent-Based Modeling

Impact of Lifestyle Factors on Students' Productivity and Mental Health

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Course – ICT 358

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Impact of Lifestyle Factors on Students' Productivity and Mental Health

1. Introduction

Nowadays students are facing increasing challenges in balancing their daily life, academic responsibilities and mental health in this fast-paced and demanding academic environment. Lifestyle factors such as academic pressure, procrastination, sleep pattern, caffeine intake, study routines and physical activity play an important role in shaping both productivity and mental health of the students. The increasing pressure to accomplish academically along with shifting lifestyle choices, has led to unhealthy behaviors that are harmful to both productivity and mental health. Students often normalize unhealthy behaviors such as sleeping for only two or three hours a day and studying all night long without sleeping, drinking too much caffeine to focus, procrastinating on essential tasks and lying in bed all day without physical activity. These kinds of behaviors raise significant concerns about their long-term effects on productivity and mental well-being.

This study aims to quantify the relationship between lifestyle habits, sleep deprivation, and student productivity through a computational modeling approach using agent-based simulation (ABM). By integrating key variables such as sleep hours, caffeine intake, study schedules, procrastination, physical activity levels, and academic pressure. Our model seeks to simulate the dynamic interactions between these factors and their collective influence on focus, task completion, emotional stability, stress management, and overall academic performance.

We investigated the impact of lifestyle factors on students' productivity and mental health through an agent-based simulation model developed using Python. By putting together various lifestyle factors such as caffeine intake, study hours, social interaction, procrastination, physical activity level and academic pressure, the model stimulates the dynamic interaction between these elements and their influence on focus, task completion, emotional stability, stress level, productivity and mental health. Apart from simulation analysis, we also leverage predictive modeling using data generated from ABM to predict how different lifestyle factors influence productivity and mental health over time. We use linear regression to train and build the model. Mean Squared Error (MSE) and R^2 score are used to evaluate the accuracy and reliability of the model.

The goal of this report is to highlight the quantifiable impact of lifestyle factors while offering insights into behavioral modifications that may optimize productivity and mental health by using computational simulation and predictive modeling. The findings may help students make informed decisions about their sleep patterns, study routines, and stress management strategies.

2. Theory Selection

In this project, we decided to use a Cognitive-Behavioral Resource Framework to explore how various lifestyle factors, such as sleep, study routines, and social interaction, can impact student productivity and mental health. This framework treats each student as an individual agent managing limited cognitive and emotional resources while interacting with their environment, including academic and social pressures.

Among these factors, sleep plays a particularly important role in how well students can focus and stay productive. When students sleep enough, they're usually more alert and can handle tasks better. But when they're sleep-deprived, it's much harder to concentrate or make good decisions. In our model, we tried to reflect this by using behavioral factors like sleep hours, caffeine intake, study hours, and procrastination. These inputs influence internal states like Focus and Tasks, which determine how focused or productive a student can be, which we track through something we called Long-Term Productivity (LP).

Mental health is another big part of the model. We noticed that emotional health often depends on things like how stressed students are, whether they're active, and how much they interact socially. So, we added inputs for academic pressure, social interaction, and physical activity. These influence emotional responses like Stress and Emotion, which together affect a student's Long-Term Mental Health (LMH).

To make this work over time, we implemented Agent-Based Modeling. This allowed us to simulate each student as an individual agent with their habits and responses. Some had good routines and stayed stable, while others struggled due to lack of sleep or too much pressure. This helped us see how behavior can lead to different outcomes for different students.

We also treated focus and emotional stability as limited resources, like how energy in a system is used and replenished. Students use their mental and emotional energy during the day and need rest or positive habits to recover. If they don't get that, their performance and well-being can drop, mirroring the real-world dynamics of human-environment systems as described in the project outline.

Through this simulation, we could see how small everyday habits, like sleep, exercise, or avoiding procrastination, can build up and shape long-term outcomes. Our simulation shows how these factors play out and could be a useful way to study student wellness in real life, too.

3. Literature Review

University students today face increasing academic demands and lifestyle pressures that directly influence their mental health and productivity. Key lifestyle factors, such as sleep quality, caffeine consumption, and physical activity, have been found to impact cognitive performance, emotional regulation, and overall well-being. As these influences are complex and often interdependent, computational approaches such as cognitive modeling and agent-based modeling provide promising tools for simulating and analyzing these dynamic relationships. This literature review explores prior research on the impact of lifestyle factors using these modeling approaches and discusses how they inform and support our project titled “Impact of Lifestyle Factors on Students’ Productivity and Mental Health using Cognitive Modeling and Agent-Based Modeling.”

3.1 First Study - Cognitive Impairment from Lifestyle Factors

Killgore (2010) conducted foundational research on the cognitive effects of sleep deprivation, identifying significant impairments in attention, working memory, and emotional stability. Similarly, Hershner and Chervin (2014) emphasized that irregular sleep among college students leads to poorer academic performance and heightened psychological distress. These findings were supported by cognitive simulation models using the ACT-R architecture, which demonstrated that reduced sleep significantly degrades decision-making performance and response time (Anderson et al., 2004). The study also highlights how internal states such as fatigue and stress influence cognitive outcomes, providing a basis for incorporating these parameters into cognitive modeling simulations

3.2 Second Study - Behavioral Patterns and Agent-Based Simulation

Ghorbani, Dignum, and Bots (2015) explored how agent-based modeling can be applied to simulate the social and behavioral dynamics related to health and lifestyle decisions. Their study focused on modeling how individual behaviors, such as diet, exercise, and sleep, spread across social groups through peer influence. By modeling individual agents with different routines, stress levels, and decision rules, they demonstrated how behavioral patterns can evolve in student communities over time. This work is particularly relevant to our project, as it validates the use of ABM to simulate the social influence and feedback mechanisms that may amplify or mitigate the effects of poor lifestyle choices

3.3 Third Study - Combining Cognitive and Agent-Based Approaches

Epstein (2006) proposed a hybrid framework that integrates cognitive models within agent-based systems, arguing that this approach better captures the complexity of real-world human behavior. His work in generative social science demonstrates how individual cognitive limitations (e.g., fatigue, anxiety) can influence group-level phenomena such as stress contagion or collaborative productivity. In educational contexts, this has been applied to simulate classroom dynamics, student dropout risk, and intervention strategies. The integration of cognitive and social layers

makes this approach ideal for studying students, whose productivity is shaped by both internal cognitive processes and external environmental pressures.

3.4 Relevance to Our Project

These studies collectively provide a strong foundation for our research. Killgore (2010) and Hershner & Chervin (2014) offer evidence of the cognitive impairments resulting from lifestyle imbalances, justifying our use of cognitive modeling to simulate individual productivity and mental health outcomes. The ABM approach from Ghorbani et al. (2015) supports our modeling of peer influence and routine propagation within student populations. Epstein's (2006) hybrid model encourages us to integrate cognitive and agent-based frameworks to better reflect the interplay between individual mental states and social environments. Together, these studies validate our methodological framework and help us construct a more realistic and impactful simulation of how lifestyle factors affect students.

3.5 Conclusion

Understanding the impact of lifestyle factors on students' mental health and productivity requires a multidisciplinary approach that considers both individual cognition and social interaction. Cognitive modeling enables the simulation of how sleep deprivation, stress, and caffeine intake affect mental processes, while agent-based modeling helps capture the diffusion of habits and the social dynamics that amplify these effects. By integrating insights from the reviewed studies, our project aims to contribute to a comprehensive and predictive model that can be used to inform policy, guide interventions, and promote healthier academic lifestyles.

4. Methodology

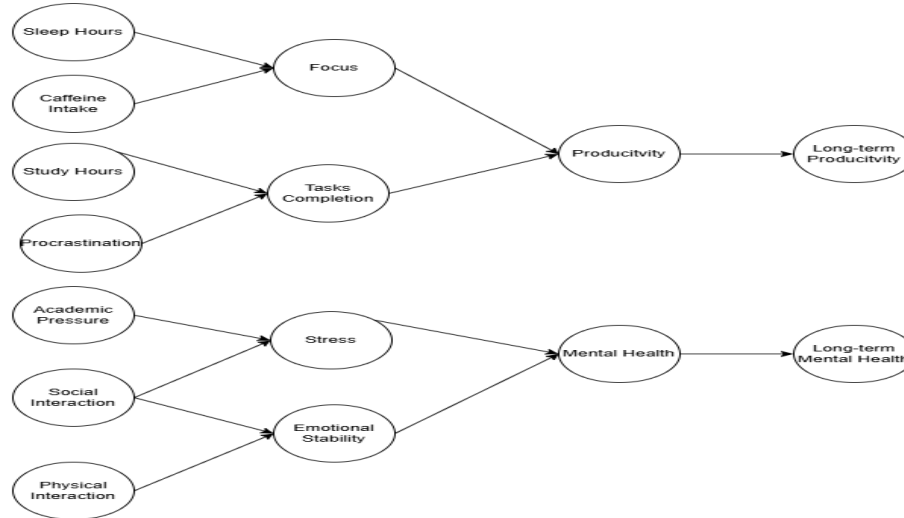
This study investigates the impact of lifestyle factors on students' long-term productivity (LP) and their long-term mental health (LMH) by integrating a computational approach using Agent-Based Modeling (ABM) combined with predictive modeling. We use the Cognitive-Behavioral Resource Framework to examine how students manage their emotional regulation while balancing schoolwork and lifestyle pressures.

We simulated two contradictory student behavior models

- High-Performance Model - Healthy behaviors (appropriate sleep, moderate caffeine intake, high study hours, low procrastination, low academic pressure)
- Low-Performance Mode - Risk behaviors (insufficient sleep, excessive caffeine intake, low study hours, high procrastination, extreme amount of academic pressure)

4.1 Cognitive Model

The cognitive model shows the process where behavioral inputs influence cognitive resource allocation, resulting in immediate outputs that shape long-term outputs (LP and LMH).



The figure above is the visual representation of the cognitive model, showing behavioral inputs, hidden cognitive processing layers and outputs.

4.2 Mathematical Representation of the Cognitive Model

The model simulates real-world student dynamics by combining behavioral elements, hidden cognitive processing layers and predictive outputs.

Inputs - Behavioral Factors

The following inputs drive cognitive and emotional processing -

- Sleep Hours (Sh)
- Caffeine Intake (C)
- Study Hours (Sth)
- Procrastination (Pr)
- Academic Pressure (Ap)
- Social Interaction (Si)
- Physical Activity (Pa)

Hidden Layers - Cognitive Processing Layers

- Focus (F)

$$F_t = \alpha \cdot Sh_t + (1 - \alpha) \cdot (1 - C_t)$$

This equation represents how sleep hours (Sh) improve focus (F), while caffeine intake (C) negatively impacts focus if excessive. The α controls the relative influence of sleep versus caffeine.

- Task Completion (T)

$$T_t = \theta \cdot Sth_t + (1 - \theta) \cdot (1 - Pr_t)$$

This equation represents how study hours (Sth) increase task completion (T), while procrastination (Pr) reduces completion rate. This factor θ determines the balance between studying and procrastination.

- Stress (S)

$$S_t = (1 - \delta) \cdot Si_t + \delta \cdot Ap_t$$

This equation shows how stress level depends on academic pressure (Ap) raising stress, while social interaction (Si) reduces stress. The parameter δ sets the sensitivity of stress to these factors.

- Emotional Stability (E)

$$E_t = \beta \cdot Si_t + \mu \cdot Pa_t$$

This equation shows emotional stability (E) improves with social interaction (Si) and physical activity (Pa), where β and μ determine their respective contributions.

Outputs - Short-Term Productivity and Mental Health

- Immediate Productivity (P)

$$P_t = \gamma \cdot F_t + \omega \cdot T_t$$

This equation represents productivity (P) is influenced by focus (F) and task efficiency (T), with γ ω weighing their effects.

- Mental Health State (MH)

$$MH_t = \sigma \cdot E_t + (1 - \sigma) \cdot (1 - S_t)$$

This equation shows mental health (MH) improves with higher emotional stability (E) but declines under high stress (S). The coefficient σ controls the balance between emotion and stress effects.

Long-Term Outputs - Long-term Productivity and Mental Health

- Long-term Productivity (LP)

$$LP_t = LP_{(t-1)} + \varepsilon_1 \cdot (P_{(t-1)} - LP_{(t-1)}) \cdot dt$$

This equation models how long-term productivity (LP) gradually adjusts over time, based on previous productivity (P), with ε_1 controlling the rate of change.

- Long-term Mental Health (LMH)

$$LMH_t = LMH_{(t-1)} + \varepsilon_2 \cdot (MH_{(t-1)} - LMH_{(t-1)}) \cdot dt$$

The above equation shows long-term mental health (LMH) evolves dynamically based on past mental health (MH), where ε_2 determines the rate of adaptation to stress and emotional states.

4.3 Data Collection and Simulation Process

Agent-Based Modeling (ABM) is used to monitor and track individual student responses to environmental conditions over time, observing their cognitive and emotional changes. Data stored includes sleep hours, caffeine intake, study hours, procrastination, academic pressure, social interaction, and physical activity. Computed cognitive metrics include focus, task completion, productivity, stress, emotional stability, and mental health. The simulation results are saved as a structured dataset (CSV format) for predictive modeling.

4.4 Regression Modeling

We used a multivariate linear regression model to predict the long-term effects of lifestyle factors on students' productivity (LP) and mental health (LMH).

- Independent Variables (X) - Behavioral inputs (Sh, C, Sth, Pr, Ap, Si, Pa) and computed cognitive layers (F, T, P, S, E, MH).
- Dependent Variables (y) - Long-term outputs (LP and LMH).

The generated dataset is divided into 80% training data and 20% is used for testing.

To evaluate the accuracy and reliability of the predictive model, the following metrics are used

- Mean Squared Error (MSE)

$$MSE = \frac{1}{n} \sum (y_{test} - y_{pred})^2$$

- R-Squared (R^2)

$$R^2 = 1 - \frac{\sum (y_{test} - y_{pred})^2}{\sum (y_{test} - y_{mean})^2}$$

5. Results and Analysis

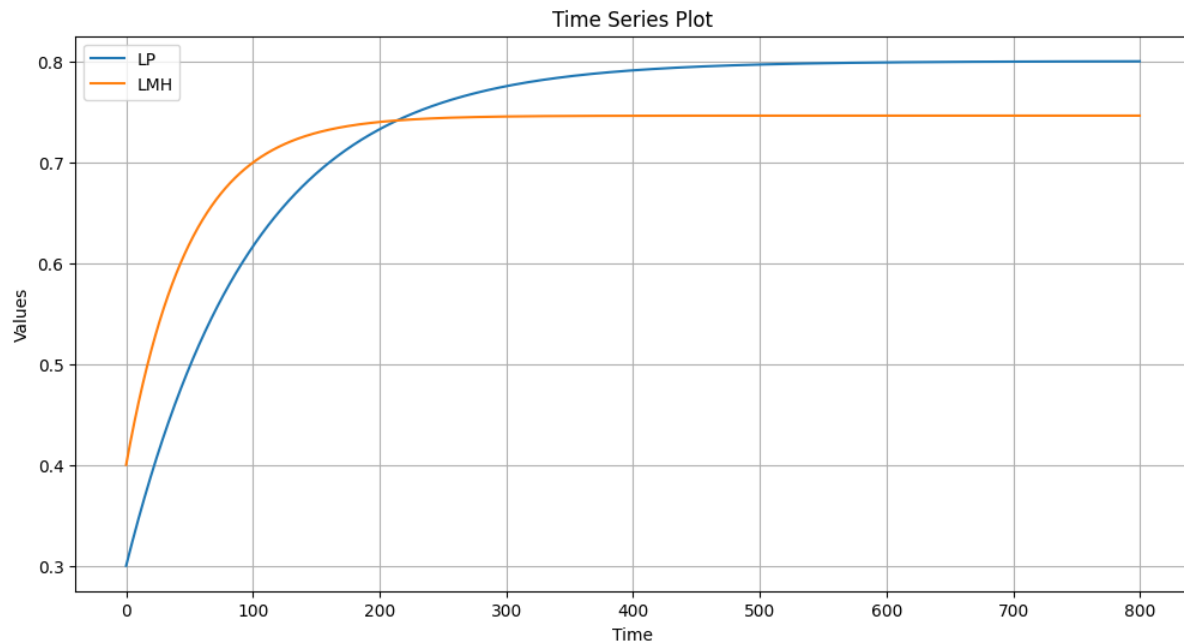
This section presents the results of our simulation comparing two different lifestyle scenarios, high performance and low performance, to assess how daily habits affect students' long-term productivity (LP) and mental health (LMH). The model was run over 800 time steps, and results were evaluated through time-series plots and regression analysis. The objective was to identify patterns in how consistent lifestyle factors shape cognitive and emotional outcomes over time.

5.1 High Performance Scenario

For the high-performance case, we used input values representing a student maintaining healthy routines, like sleep and study hours, were set to 0.8, while caffeine use and procrastination were kept low at 0.2. Academic pressure was minimal (0.2), and both physical activity and social interaction were high (0.8).

As expected, both LP and LMH gradually improved. LP started at 0.3 and climbed to around 0.8. LMH, which began at 0.4, stabilized near 0.75. This confirmed that when students consistently follow healthy routines, their focus and emotional well-being improve over time.

We also ran a linear regression model to check how well the input values could predict LP and LMH. The model showed the squared error of 0.00298 and R^2 of 0.4608. While the error was slightly higher than in the low-performance case, the predictive strength remained consistent.



5.2 Low Performance Scenario

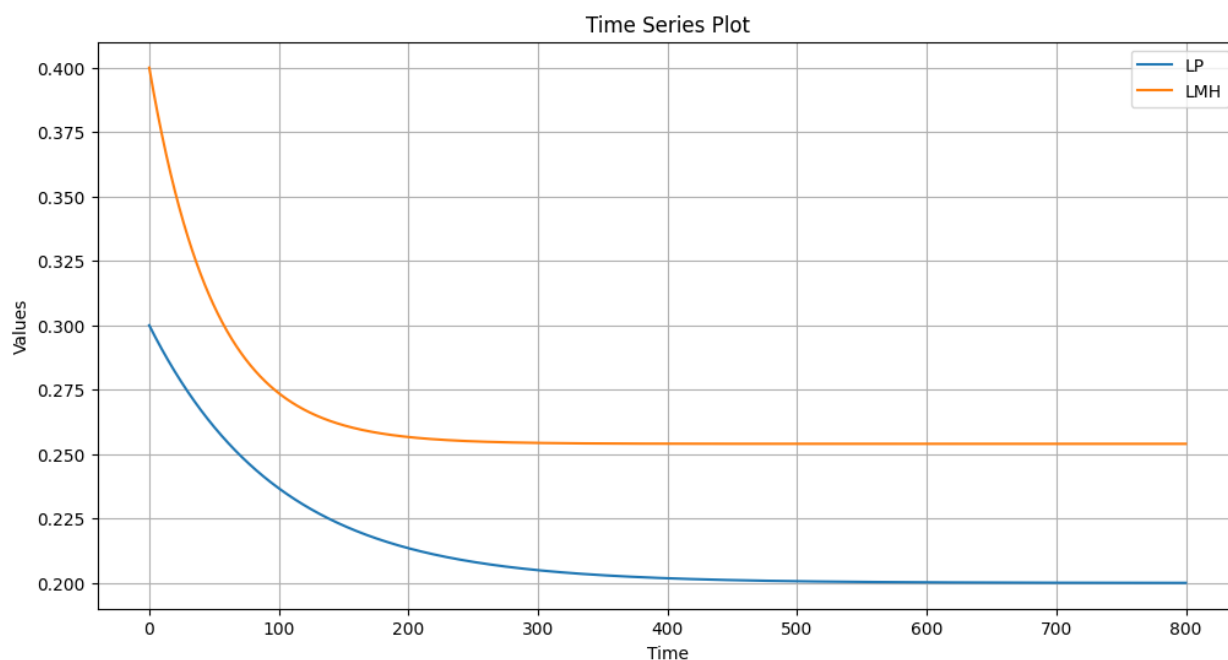
In contrast, we simulated a low-performance scenario by reversing the inputs. Sleep and study were reduced to 0.2, caffeine and procrastination were raised to 0.8, academic pressure was high (0.8), and both social interaction and physical activity were minimized to 0.2.

The result showed a steady decline in both LP and LMH. LP dropped toward 0.2, and LMH eventually settled just above that, around 0.25. These results were consistent with our assumption that a lack of healthy habits can steadily weaken both mental well-being and academic performance.

The regression results in this scenario were

- Mean Squared Error – 0.00023
- R-squared – 0.4608

Although the MSE was smaller here, the R^2 value stayed the same. This consistency supports the idea that the model responds reliably even across very different inputs.



5.3 Comparative Insights

Comparing both cases helped us see how lifestyle factors can make a noticeable difference over time. Even though both simulations ran with the same structure, the outcome curves went in opposite directions. We also noticed that LMH responded faster than LP in most cases, which may suggest emotions are often more reactive than long-term performance.

In general, this project showed how agent-based modeling can help us visualize how small, repeated choices shape student outcomes. It also demonstrated how the balance between effort, recovery, and environment affects both learning and well-being.

6. Conclusion

Through this project, we explored the impact of various lifestyle factors, such as sleep duration, caffeine consumption, procrastination, and physical activity, on students' productivity and mental health. By applying multivariate linear regression, we identified key relationships between these variables and student outcomes. The regression model revealed that sufficient sleep and regular physical activity had a strong positive correlation with productivity and mental well-being, while high caffeine intake and excessive procrastination were negatively associated with performance. Additionally, agent-based modeling allowed us to simulate diverse student behaviors in a controlled environment, helping us observe how lifestyle choices evolve over time and affect long-term productivity. The analysis demonstrated that students who maintained balanced habits consistently outperformed those with irregular or unhealthy routines. These insights emphasize the importance of encouraging students to adopt healthier daily practices to enhance both academic success and psychological resilience. Our findings not only contribute to existing literature but also offer practical guidance for educational institutions aiming to support student wellness through evidence-based strategies.

References

- Anderson, J. R., Bothell, D., Byrne, M. D., Douglass, S., Lebiere, C., & Qin, Y. (2004). An integrated theory of the mind. *Psychological Review*, 111(4), 1036–1060. <https://doi.org/10.1037/0033-295X.111.4.1036>
- Epstein, J. M. (2006). *Generative Social Science: Studies in Agent-Based Computational Modeling*. Princeton University Press. <https://www.jstor.org/stable/j.ctt7rxj1>
- Ghorbani, A., Dignum, V., & Bots, P. W. G. (2015). Agent-based modeling for simulation of smart societies: A literature review. 2015 Winter Simulation Conference (WSC), 2788–2799. <https://doi.org/10.1109/WSC.2015.7408344>
- Hershner, S. D., & Chervin, R. D. (2014). Causes and consequences of sleepiness among college students. *Nature and Science of Sleep*, 6, 73–84. <https://doi.org/10.2147/NSS.S62907>
- Killgore, W. D. S. (2010). Effects of sleep deprivation on cognition. *Progress in Brain Research*, 185, 105–129. <https://www.scirp.org/reference/referencespapers?referenceid=3920063>
- Elmeligy Abdelhamid, S. &. (2016). Agent-Based Modeling and Simulation of Depression and Its Impact on Student Success and Academic Retention. New Orleans, Louisiana: ASEE Annual Conference & Exposition. Retrieved from <https://peer.asee.org/26545>
- Glenn Gunzelmann, J. B. (2009). Sleep Deprivation and Sustained Attention Performance: Integrating Mathematical and Cognitive Modeling. *Cognitive Science*. Retrieved from <https://doi.org/10.1111/j.1551-6709.2009.01032.x>